PYGEOVISION · GEOSPATIAL ANALYSIS REPORT

Urban Expansion (annual 10 m land cover)

2020 to 2023 · 2.8 km²

172.1 ha	177.7 ha	16.8 ha	11.2 ha
Built area, first year	Built area, second year	Gross gain	Gross loss

Executive summary

- Built-up area went from 172.1 ha in 2020 to 177.7 ha in 2023: gross gain 16.8 ha, gross loss 11.2 ha, net 5.6 ha.
- Compound annual growth of the built-up area: 1.1 % per year.

Impact Observatory / Esri 10 m Annual Land Use Land Cover (v02, Sentinel-2 based, CC BY 4.0). Each year is classified independently, so a year-to-year difference contains classifier disagreement as well as real change; single-pixel flips are expected.

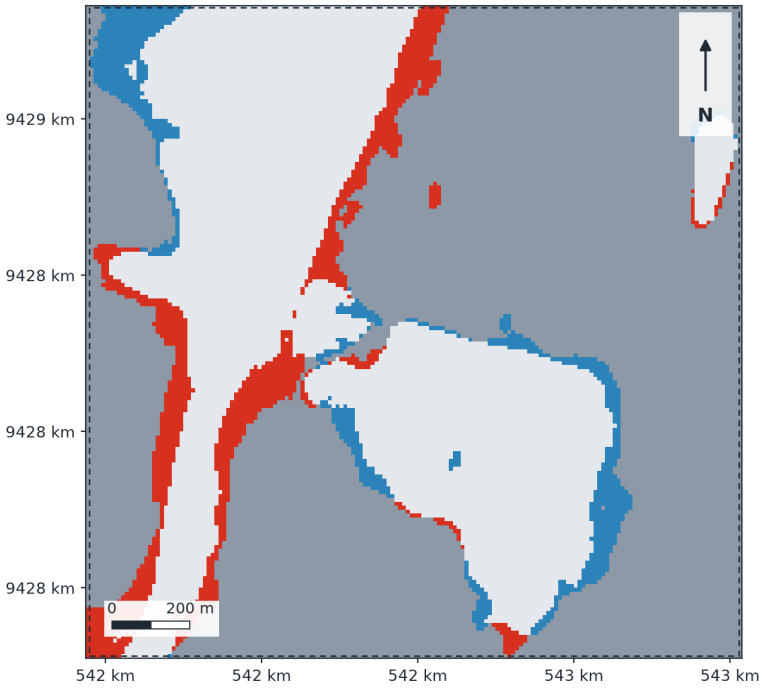
Study area and inputs

Area of interest (west, south, east, north)	-80.6250, -5.1800, -80.6100, -5.1650
Centre	-5.1725°, -80.6175°
Area of interest	2.76 km²
Period	2020 to 2023
Output grid	168 × 167 pixels, EPSG:32717
Raster extent area	2.81 km²
Valid (cloud-free) share	100.0 %

Results

Urban Expansion (annual 10 m land cover)

Built-up area change · 2020 to 2023 · AOI -80.6250, -5.1800, -80.6100, -5.1650



Legend

- **Not built** — 91.7 ha · 32.7 %
- **Built in both years** — 160.9 ha · 57.3 %
- **New built area** — 16.8 ha · 6.0 %
- **Built area lost** — 11.2 ha · 4.0 %

Key figures

Built area, first year
172.1 ha

Built area, second year
177.7 ha

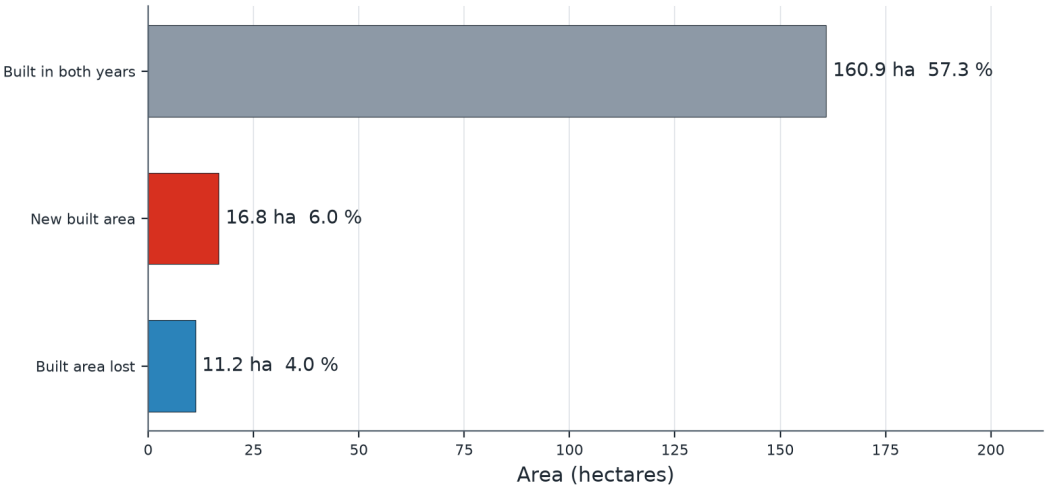
Gross gain
16.8 ha

Gross loss
11.2 ha

CRS: EPSG:32717 · Areas measured from the output raster · No basemap · PyGeoVision 2.2.0 · 2026-10-30 07:15 UTC

Map: Urban Expansion (annual 10 m land cover)

Area by class: Built-up area change



Summary chart

Class areas: Built-up area change

Class	Value	Pixels	Area (ha)	Share of valid area	Share of AOI
Not built	0	9,173	91.7	32.7 %	32.7 %
Built in both years	1	16,089	160.9	57.3 %	57.3 %
New built area	2	1,677	16.8	6.0 %	6.0 %
Built area lost	3	1,117	11.2	4.0 %	4.0 %

Valid area 280.6 ha of 280.6 ha (28,056 of 28,056 pixels). Areas are measured from the output raster.

Key statistics (reported by the pipeline)

Measure	Value
Built area, first year	172.1 ha
Built area, second year	177.7 ha
Gross gain	16.8 ha
Gross loss	11.2 ha
Net change	5.6 ha
Built share, first year	61.3 %
Built share, second year	63.3 %
Compound annual growth of built area	1.1 %
Valid share of the area	100.0 %

Other statistics

Measure	Value
Year before	2,020
Year after	2,023
Resolution m	10.000
Resolution coarsened	no

Method and parameters

How much built-up area appeared or disappeared between two years?

Built area is class 7 of the annual 10 m land-cover product (Impact Observatory / Esri, 2017-2023) on both dates.

A pixel is 'new built' when it is built in the second year only, 'lost' when built in the first year only. Gross gain and loss are reported separately because each year is classified on its own.

Pixels that are no-data or cloud in either year are nodata; percentages are over valid pixels.

Pipeline	urban_expansion
Area of interest	-80.6250, -5.1800, -80.6100, -5.1650
date_after	2023
date_before	2020
period	2020 to 2023

Interpretation and limitations

How to read the result

- 'Built area lost' is more often classifier disagreement (roofs under trees, dust, construction sites) than demolition. Read the gross figures, not only the net.

Limitations

- At 10 m, individual houses and narrow streets are not resolved; small settlements are under-mapped.
- This is a map product's change, not a validated change inventory.

Deliverables

Files

File	Type	Contents
piura_out/urban_expansion/built_area_change.tif	GeoTIFF raster	Built-up area change
piura_out/urban_expansion/deliverables/urban_expansion_report.pdf	PDF report	Complete analysis report
piura_out/urban_expansion/deliverables/urban_expansion_report.xlsx	Excel workbook	Complete analysis report
piura_out/urban_expansion/deliverables/urban_expansion_report.docx	Word report	Complete analysis report
piura_out/urban_expansion/deliverables/maps/urban_expansion_map.png	PNG image	Map or chart for slides and documents
piura_out/urban_expansion/deliverables/maps/urban_expansion_chart.png	PNG image	Map or chart for slides and documents
piura_out/urban_expansion/deliverables/vectors/urban_expansion.gpkg	GeoPackage (all layers)	Vector features with attributes
piura_out/urban_expansion/deliverables/vectors/geojson/urban_expansion_aoi.geojson	GeoJSON, EPSG:4326	Vector features with attributes
run_summary.json	JSON	Machine-readable record of this run

References

- Karra, K. et al. (2021). Global land use / land cover with Sentinel-2 and deep learning. IGARSS 2021, 4704-4707. (Impact Observatory / Esri 10 m annual land cover, io-lulc-annual-v02, Microsoft Planetary Computer.)

Notes on this run

Parts of the deliverables that could not be produced, or were changed, and why:

- vector layer 'built_change' could not be created: No module named 'scipy'